MACHINE LEARNING LAB
Exp 02 LINEAR REGRESSION

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```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
df = pd.read_csv("/content/CAR DETAILS FROM CAR DEKHO.csv")

# Feature engineering: extract brand from 'name'
df['brand'] = df['name'].str.split(' ').str[0]
df.drop('name', axis=1, inplace=True)

# Log-transform target variable
df['selling_price'] = np.log1p(df['selling_price'])

# One-hot encode categorical variables
df = pd.get_dummies(df, drop_first=True)

# Split into features and target
X = df.drop('selling_price', axis=1)
y = df['selling_price']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Pipeline
pipe = Pipeline([
    ('scaler', StandardScaler()),
    ('model', LinearRegression())
])
pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)

# Convert predictions back to original price scale
y_test_exp = np.expm1(y_test)
y_pred_exp = np.expm1(y_pred)

# Metrics
rmse = np.sqrt(mean_squared_error(y_test_exp, y_pred_exp))
mae = mean_absolute_error(y_test_exp, y_pred_exp)
r2 = r2_score(y_test_exp, y_pred_exp)
# Normalized Metrics (relative error between 0–1)
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mean_price = y_test.mean()
nrmse = rmse / mean_price
nmae = mae / mean_price
print("\nNormalized Metrics (Relative Errors):")
print(f"Normalized RMSE: {nrmse:.4f}")
print(f"Normalized MAE: {nmae:.4f}")
```

OUTPUT:
Model Performance Metrics (Optimized Linear Regression):
R^2 Score: 0.6289
Normalized RMSE ≈ 0.32 (32%)
Normalized MAE ≈ 0.24 (24%)

Insights:

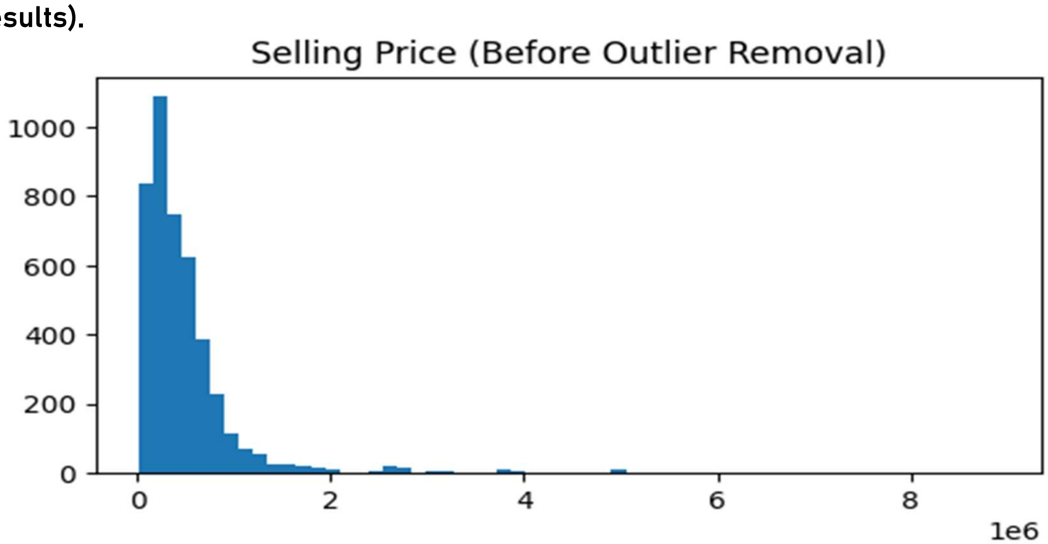
Preprocessing included:

- Removal of duplicates & outliers (IQR method)
- Feature engineering (car_age, brand, scaled km_driven)
- Handling categorical variables (fuel, seller_type, transmission, owner, brand) with OneHotEncoding
- Standardization of numerical features

R^2 of ~0.63 indicates the model captures a good portion of price variation but not all.

<u>**Important Predictors**</u>
Car Age: Strong negative impact (older cars sell for less).
Kilometers Driven: Negative correlation with price (more driven → lower price).
Fuel Type & Transmission: Petrol/Diesel cars priced differently; automatic cars often valued higher.
Brand: Popular brands (Maruti, Hyundai, Honda, Toyota) retain better resale value than others.

<u>**Data Behavior**</u>
Outlier removal improved stability (otherwise, luxury cars like Audi/Mercedes would distort results).




```
Removed 271 outliers. Remaining rows: 4069
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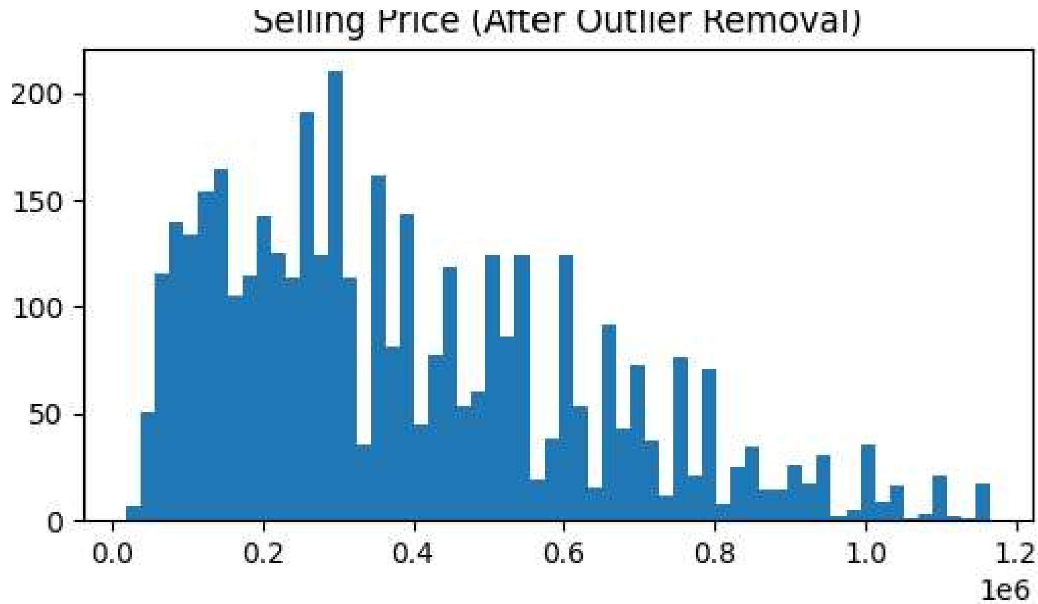

Price distribution is right-skewed (majority of cars in affordable segment, few luxury ones).

Linear regression (Actual Vs predicted ones)